University of Tunis
Tunis Business School

Fall 2023

Production & Operation Management BA 305
Midterm Exam
(Duration 2 hours)

Question 1

1. list the essential functions of a company and their roles.
2. What is the purpose of the Jackson's rule? Briefly explain the main steps of this algorithm.
3. What do we mean by production, operation management, and productivity ?

Question 2

In the final report of academic years 2021-2022 and 2022-2023 of the Business Analytics and Decision-Making lab at TBS, we find the following table:

	2021-2022	2022-2023
Budget in TND	45,000	55,000
Number of researchers	16	14
Number of research papers produced	14	15
Number of research papers accepted for publication in top international journals	6	5

1. Identify the lab inputs and outputs used in the above table.
2. Identify the percent change in productivity with respect to various combinations of input/output.
3. Can we say that by further reducing the number of researchers, the productivity with respect to the papers production is expected to increase?

Question 3

We want to sequence the following five jobs on a machine.

Job	Processing Time	Due Date
A	39	65
B	2	3
C	52	52
D	23	44
E	49	49

What is the optimal order of tasks to minimize the total completion time? Justify your answer.

Question4

Consider the list of tasks along with their execution times (in hours) in centers 1 and 2.

Task	Center 1	Center 2
A	4	6
B	6	2
C	2	6
D	10	6

1. Use the Johnson algorithm to schedule these tasks in both centers.
2. Draw the Gantt chart and deduce the minimal time required to complete all tasks.
3. Determine the idle time for Center 2.
4. Suppose that each of tasks C and D can be split into two subtasks of equal duration. Draw the corresponding Gantt chart and deduce the effect of this splitting on Center 1, on the idle time of Center 2, and on the total processing time of the tasks.